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Research Paper

A fully coupled three-dimensional hydro-mechanical finite discrete element approach with real porous seepage for simulating 3D hydraulic fracturing

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ABSTRACT

A fully coupled 3D hydro-mechanical model with real porous seepage is presented for simulating hydraulic fracturing. In this model, fluid flow in a fracture is expressed by 2D fracture seepage in the broken joint elements based on the Cubic law, while fluid flow in the rock matrix is represented by 3D porous seepage in the tetrahedral elements based on Darcy's law. Several problems that have closed-form solutions and a 3D fracturing problem are given to verify the model. The simulation results show that the model can capture crack initiation and propagation, and the fluid pressure evolution during hydraulic fracturing.

1. Introduction

Hydraulic fracturing is the key technology in petroleum, shale gas and enhanced geothermal systems. It involves not only the deformation of a solid under the effect of fluid pressure but also the fracturing of the solid. Many researchers have studied hydraulic fracturing and put forward several theoretical models, such as the PKN model [1,2], the KGD model [3,4], the radial or penny-shaped model [5], and some pseudo-3D models and planar-3D models [6–8]. Later, rather than developing new analytical models for hydraulic fracturing with complex fracture networks, researchers have been focused on the properties of classical hydraulic fracturing models. For example, a scaling and asymptotic framework was developed by Detournay et al. [9,10], who recognized that the hydraulic fracture is governed by two competing energy (i.e., viscous flow and the creation of surface area in the solid) and fluid storage mechanisms (i.e., the storage of fluid in the fracture and fluid leak-off into the permeable solid). Later, many fruitful semi-analytical and numerical solutions have been obtained for different asymptotic regimes [11–18], such as zero toughness impermeable, small toughness impermeable, finite toughness impermeable, large toughness impermeable, zero toughness permeable regime, and finite toughness permeable regime solutions [19]. Although these theoretical models or solutions are only applicable to simple geometries that are very different from an actual fracturing scenario with complex natural fracture networks, they are important in understanding hydraulic fracturing and provide benchmarks for numerical simulations.

Many numerical models have been also used to study hydraulic fracturing, such as the finite element method (FEM), the extended finite element method (XFEM), the displacement discontinuity method (DDM), the discontinuous deformation analysis method (DDA), the discrete element method (DEM), and the finite discrete element method (FDEM).

For example, Fu et al. [20] and Settgast et al. [21] built an explicit 2D/3D hydraulic coupling model to simulate hydraulic fracturing with arbitrary fracture networks based on FEM. Hunsweck et al. [22] presented a finite element approach to simulate plane-strain hydraulic fractures with lag. Gupta and Duarte [23] presented simulations of non-planar three-dimensional hydraulic fracture propagation. Chen [24] proposed a 3D finite element model based on existing pore pressure cohesive finite elements to simulate the propagation of viscosity-dominated hydraulic fracture in an infinite, impermeable elastic medium. Wangen [25] suggested a finite element approach for the modeling of hydraulic fracturing in 3D. Yao et al. [26] developed a 3D pore pressure cohesive zone model for simulation of hydraulic fracturing in quasi-brittle rocks. However, FEM needs remeshing (with high computation cost) as fractures continually propagate [27–29].

In order to overcome the limitations of the finite element method, some researchers have used XFEM to simulate hydraulic fracturing. For example, Salimzadeh and Khalili [36] proposed a fully coupled three-phase model for simulating hydraulic fracturing in porous media, where XFEM is used to handle discontinuities and a cohesive crack model is used as a fracturing criterion. Faivre et al. [37] proposed a 2D coupled HM-XFEM with cohesive zone model and applied it to a fluid-driven fracture. Mohammadnejad and Khoei [38] developed a fully coupled

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numerical model for the modeling of hydraulic fracture propagation in porous media using XFEM in conjunction with the cohesive crack model. XFEM can avoid the remeshing, but it needs to discretize the solution domain into a fine mesh if the crack propagation path is unknown in advance, which will lead to a very large computational load, and cannot effectively simulate complex multiple-crack propagation and intersection.

Some researchers have used DDM to simulate hydraulic fracturing [39,40]. For example, Kresse et al. [41] used DDM to study fluid-driven ruptures in natural reservoirs, but the rock matrix was assumed to be impermeable. Norbeck [42] introduced an efficient numerical modeling framework that integrated the embedded fracture modeling (EFM) strategy into a DDM to simulate hydraulic fracturing. Ganis et al. [43] proposed an algorithm for modeling saturated fractures in a poroelastic domain in which the reservoir simulator is coupled with DDM. The simulation of hydraulic fracturing with DDM usually assumes that the rock matrix is impermeable and cannot consider fluid infiltration into the rock matrix from the fracture. In addition, these simulations are usually limited to two-dimensional cases.

When dealing with discontinuous problems, especially with a large number of natural fractures, those continuum numerical methods have some inherent flaws. For this reason, some researchers have used discontinuous numerical methods (e.g., DDA, DEM) to study hydraulic fracturing. For example, DDA was used to simulate hydraulic fracturing by taking the hydro-mechanical coupling and the fragmentation of the block into account [44–47]. Some researchers used the discrete element method to study hydraulic fracturing. For example, Nasehi and Mortazavi [48] and Hamidi and Mortazavi [49] introduced virtual joint technology in UDEC/3DEC to simulate crack initiation and propagation of a typical reservoir driven by fluid. Although these methods can simulate crack initiation and propagation, the fluid flow in fractures and the interactions between fluids and solids, they do not take into account fluid leakage into the surrounding rock matrix from fractures, i.e., the permeability of the rock matrix is assumed to be zero. The particle flow method [50] is another method that can be used to simulate hydraulic fracturing [51–57]. However, the microscopic parameters of the method are difficult to calibrate, and the fracture characterization is not intuitive. In addition, the concept of stress and strain in continuum mechanics no longer directly exists in this method.

To overcome the shortcomings of the above methods in simulating hydraulic fracturing, we construct a fully coupled 3D hydro-mechanical model with real porous seepage in 3D FDEM in this paper. In this model, the porous seepage in the rock matrix is represented by a real three-dimensional flow, rather than the equivalent approach [34,68,69] that the porous seepage in the rock matrix is represented by the fracture seepage in the unbroken joint elements. For an equivalent approach, it is necessary to calibrate the initial aperture of the unbroken joint element to characterize the macroscopic permeability of the rock sample. In addition, the equivalent approach has the lag phenomenon in fluid flow when dealing with unsteady porous seepage and cannot consider the pores in rock. Thus, the equivalent method cannot deal well with the unsteady problem of hydraulic fracturing. However, the fully coupled 3D model described in this paper does not contain these shortcomings; it can consider the permeability anisotropy of the rock matrix and the fluid loss from the fracture into the rock matrix. Furthermore, the fully coupled 3D model can simulate crack initiation and propagation, and the interaction between fluid and solid. Since cracks extend along the tetrahedral element boundary, remeshing is not needed. The cracks consist of triangular faces, and the characterization of the cracks is very intuitive. FEM is used to calculate the stress and strain of the tetrahedral elements, and thus the concept of stress and strain in continuum mechanics is well preserved in FDEM. In addition, the discrete element method is used to process contacts between elements in FDEM; the contacts can be easily handled when the crack is closed.

This article is organized as follows. First, the fundamentals of 3D FDEM are briefly introduced, including the governing equation, contact detection, contact force and the joint element constitutive model. Next, the basic assumptions of the three dimensional fully coupled model are introduced. The 3D real porous seepage model and fracture seepage model is presented in Section 4. Then, the fracture-stress-seepage coupling process is introduced in Section 5. Finally, the fully coupled model is thoroughly verified using two examples that have analytical solutions. The third example is a hydraulic fracturing problem with complex fracture networks in which the interaction between hydraulic fracturing and pre-existing fractures is studied. Moreover, the effect of in situ stress on hydraulic fracturing is investigated in the last example and compared with experimental result.

2. Fundamentals of the 3D finite discrete element method

The finite discrete element method (FDEM) [58,59] is an excellent method for simulating fracture of solid material. In recent years, FDEM has been widely used in the field of rock mechanics, especially for problems related to rock fracture [60–63]. Some of the literature based on FDEM and related to this paper can be found in Latham et al. [64], Lei et al. [65,66], Grasselli et al. [67], Yan et al. [30,31,33,35,68], and Lisjak et al. [69].

In 3D FDEM, each individual block is discretized into a finite element mesh with tetrahedral elements, and a 6-node joint element with zero initial thickness is inserted on the common surface between the adjacent tetrahedral elements, as shown in Fig. 1. The finite element method is used to determine the stress and strain of the constant strain tetrahedral element, while the discrete element method is used to deal with the contact between the tetrahedral elements so that the method can simulate the sliding and rotation of the block. In addition, the breakage and fragmentation of the block can be modeled by the joint element breaking. The deformation of the block can be simulated by a deformable tetrahedral element and the unbroken joint element with a bonding effect. Next, we describe the governing equations, contact detection and contact force, and the joint element constitutive model.

2.1. Governing equations

The governing equations of FDEM are essentially the same as those of DEM, that is, according to Newton's second law, the dynamic equations of the nodes are given by

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}), \tag{1}$$

where $\mathbf{M}$ and $\mathbf{C}$ are the mass and damping diagonal matrices of all of the nodes in the system, $\mathbf{F}(\mathbf{x})$ denotes the unbalanced nodal force vector. It includes the nodal force $\mathbf{F}_c$ caused by the contact force, the nodal force $\mathbf{F}_d$ caused by the deformation of the tetrahedral elements, the nodal force $\mathbf{F}_e$ caused by the external load, and the nodal force $\mathbf{F}_j$ caused by the bonding stress of the joint elements. The damping matrix $\mathbf{C}$ is used to dissipate the kinetic energy of the system when solving the static problems by dynamic relaxation. The damping matrix $\mathbf{C}$ is given by

$$\mathbf{C} = \mu\mathbf{I}, \tag{2}$$

where μ is the damping coefficient and $\mathbf{I}$ is the unit matrix. According to the single-degree-of-freedom mass spring system, the critical damping coefficient is given by [59]

$$\mu = 2h\sqrt{\rho E}, \tag{3}$$

where h is the length of the element, ρ is the density of the solid, and E is the elastic modulus. When critical damping is used, the kinetic energy of the system can be consumed with the fastest speed.

The nodal force $\mathbf{F}_d$ caused by the deformation of tetrahedral elements can be obtained directly from the linear elastic constitutive law, while the nodal force $\mathbf{F}_c$ caused by the contact force and the nodal force $\mathbf{F}_j$ caused by the bonding stress of joint elements will be presented in Section 2.2 and Section 2.3 in detail, respectively.

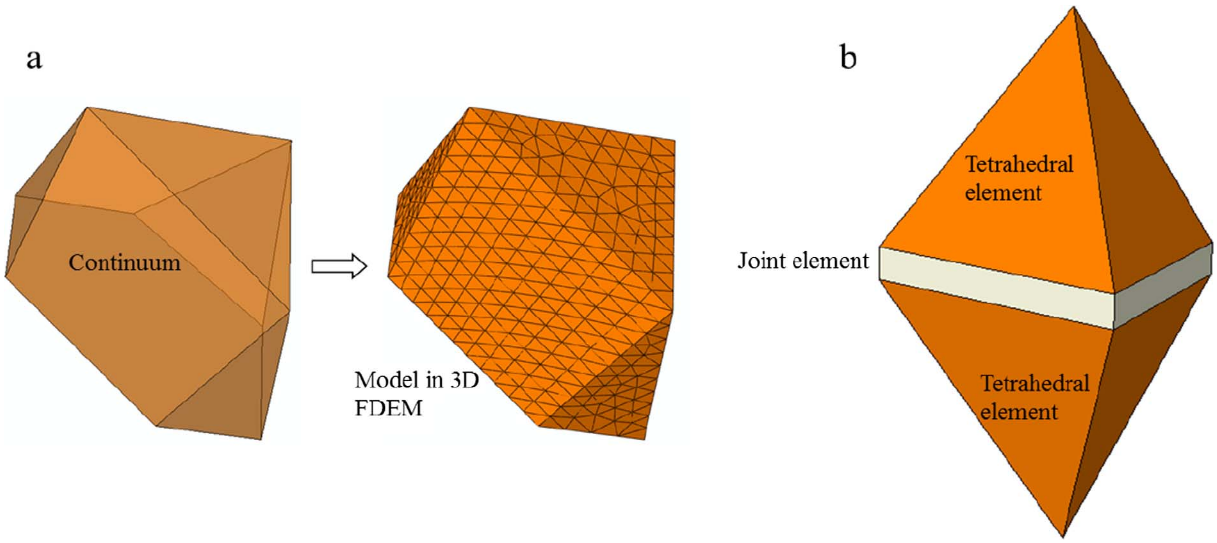


Fig. 1. Connection of tetrahedral elements and joint elements in 3D FDEM.

2.2. Contact detection and contact force

2.2.1. Contact detection

In FDEM, the NBS [70] contact detection algorithm is used to find the potential contact pairs. The contact detection time is a linear function of the number of elements when the elements are of similar size. Later, Munjiza further developed the algorithm and proposed the MR [71] contact detection algorithm. The algorithm can also efficiently deal with contact detection in systems with large differences in element size. Using the above algorithm, the contact pairs of tetrahedral elements are found. Then, the normal contact force is calculated by the potential method while the tangential contact force is calculated based on the Coulomb friction, as shown in the following section.

2.2.2. Contact force

The contact force consists of two parts, the normal contact force and the tangential contact force. The normal contact force is calculated by the potential method [72], which can ensure energy conservation during a frictionless collision. The tangential contact force is calculated based on classical Coulomb friction.

As shown in Fig. 2, one contact pair contains two tetrahedral elements in contact, one of which is denoted as the contactor and the other as the target [59]. The contact force of the contactor embedded in the target with volume $\mathrm{d}V$ is given by

$$\mathrm{d}\boldsymbol{f}_{c1} = -p(\mathbf{grad}\varphi_t(P_t))\mathrm{d}V, \tag{4}$$

where p is the normal penalty, P_t is a point of the target β_t in the overlapping part, and $\varphi_t(P_t)$ is the potential of the point P_t in the target β_t .

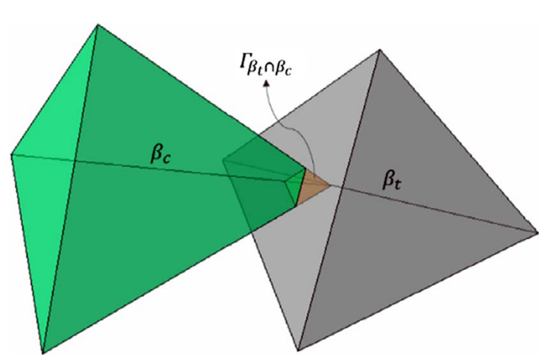


Fig. 2. Schematic of potential contact force calculation.

Similarly, the contact force of the target embedded in the contactor with volume $\mathrm{d}V$ is given by

$$\mathrm{d}\boldsymbol{f}_t = -p(\mathbf{grad}\varphi_c(P_c))\mathrm{d}V. \tag{5}$$

It is important to note that the contact force calculated by Eq. (5) acts on the target through the contactor which provides the potential. Thus, the reaction force of the target to the contactor is equal to the magnitude of Eq. (5) but in the opposite direction

$$\mathrm{d}\boldsymbol{f}_{c2} = p(\mathbf{grad}\varphi_c(P_c))\mathrm{d}V. \tag{6}$$

Thus, the total contact force (target to contactor) due to the overlapping volume $\mathrm{d}V$ is given by

$$\mathrm{d}\boldsymbol{f} = df_{c1} + df_{c2} = p[\mathbf{grad}\varphi_c(P_c) - \mathbf{grad}\varphi_t(P_t)]\mathrm{d}V. \tag{7}$$

By integration of Eq. (7) over the entire overlapping volume V, the total contact force between the contact pair can be given by

$$\boldsymbol{f} = \int_{V=\beta_t\cap\beta_c} p[\mathbf{grad}\varphi_c(P_c) - \mathbf{grad}\varphi_t(P_t)]\mathrm{d}V. \tag{8}$$

According to the Gaussian formula, Eq. (8) can be written as an integration over the outer surface of the overlapping zone

$$\boldsymbol{f} = p\int_{S_{\beta_t\cap\beta_c}} \boldsymbol{n}(\varphi_c(P_c) - \varphi_t(P_t))\mathrm{d}S, \tag{9}$$

where $\boldsymbol{n}$ is the outer normal unit vector of the surface of the overlapping zone V.

The individual discrete blocks are discretized into tetrahedral finite elements in 3D FDEM. Thus, each discrete block can be represented by the union of those tetrahedral elements

$$\left.\begin{array}{l}\beta_c = \beta_{c_1} \cup \beta_{c_2} . .. \cup\beta_{c_i} . .. \cup\beta_{c_n}\\ \beta_t = \beta_{t_1} \cup \beta_{t_2} . .. \cup\beta_{t_j} . .. \cup\beta_{t_m}\end{array}\right\}. \tag{10}$$

The volume integral of the overlapping zone of the discrete blocks can then be expressed as the sum of the integration over the overlapping zone of the tetrahedral elements [59]

$$\boldsymbol{f} = \sum_{i=1}^{n}\sum_{j=1}^{m}\int_{\beta_{c_i}\cap\beta_{t_j}} p[\mathbf{grad}\varphi_{c_i}(P_{c_j}) - \mathbf{grad}\varphi_{t_j}(P_{t_j})]\mathrm{d}V. \tag{11}$$

According to the Gaussian formula, the volume integral of the overlapping zone is converted to the surface integral of the boundary of the overlapping zone

$$\boldsymbol{f} = \sum_{i=1}^{n} \sum_{j=1}^{m} p \int_{S_{\beta_{c_i} \cap \beta_{t_j}}} \boldsymbol{n}(\varphi_{c_i} - \varphi_{t_j})]\mathrm{d}S. \tag{12}$$

After obtaining the normal contact force $\boldsymbol{f}$ and its acting point, we calculate the tangential contact force according to classical Coulomb friction.

$$\boldsymbol{f}_{\mathrm{t}}^{t} = \boldsymbol{f}_{\mathrm{t}}^{t-\Delta t} - p_{\mathrm{t}} \Delta \mathbf{u}_{s} S_{B} \tag{13}$$

where $\boldsymbol{f}_{\mathrm{t}}^{t}$ is the test value at the current time step, $\boldsymbol{f}_{\mathrm{t}}^{t-\Delta t}$ is the friction at the previous step, p_{t} is the tangential penalty, and $\Delta \mathbf{u}_{s}$ is the tangential relative displacement increment at the current time step.

If the test value $|\boldsymbol{f}_{\mathrm{t}}^{t}| \leqslant u|\boldsymbol{f}|$ (u is friction coefficient) calculated by Eq. (13), then

$$\boldsymbol{f}_{\mathrm{t}} = \boldsymbol{f}_{\mathrm{t}}^{t}. \tag{14}$$

If the test value $|\boldsymbol{f}_{\mathrm{t}}^{t}| > u|\boldsymbol{f}|$ calculated by Eq. (13), then

$$\boldsymbol{f}_{\mathrm{t}} = \frac{\boldsymbol{f}_{\mathrm{t}}^{t}}{|\boldsymbol{f}_{\mathrm{t}}^{t}|} u|\boldsymbol{f}|. \tag{15}$$

According to Eqs. (14) and (15), we obtain the tangential contact force.

2.3. Constitutive model of the joint element

Since the fracturing in the 3D FDEM is achieved by the joint element breaking, the constitutive behavior of the joint element is of critical importance for crack initiation and propagation. A joint element connects two surfaces of two neighboring tetrahedral elements. According to the relative displacement of the two surfaces, the joint element may be in the elastic state (peak front), the yield state (after the peak) or the failure state (i.e., the joint element breaks). There are three types of failure, Mode I (tensile failure), Mode II (shear failure), and mixed Mode I-II (tensile-shear mixed failure).

As shown in Fig. 3a, when the normal opening amount o of two surfaces connected by a joint element reaches the critical value o_p, the normal bond stress σ of the joint element is exactly equal to the tensile strength f_t. If the normal opening amount o continues to increase, the normal bond stress σ is gradually reduced until it reaches the maximum normal opening amount o_r at which point the normal bond stress is zero and a tensile crack is generated.

As shown in Fig. 3b, when the tangential slipping amount s of the two surfaces connected by the joint element reaches the critical value s_p, the tangential bond stress τ of the joint element is just equal to the shear strength f_s. If the tangential slipping amount s continues to increase, the tangential bond stress τ gradually decreases until the maximum tangential slipping amount s_r is reached. At this time, the joint element breaks and no tangential bond stress exists. The shear strength is determined according to the Mohr-Coulomb criterion

$$f_s = c - \sigma \tan\phi, \tag{16}$$

where c is the cohesion, σ is the normal stress of the joint element, which has a negative sign because tensile stress is positive, and ϕ is the internal friction angle.

In addition to the two failure modes previously discussed, there is another failure mode: the mixed mode I-II (tensile-shear mixing failure), as shown in Fig. 3c. In this mode, the normal opening amount and the tangential slipping amount are smaller than the maximum normal opening amount o_r and the maximum tangential slipping amount s_r, respectively, but the two surfaces connected by the joint element are both in normal opening and tangential slipping and satisfy the following condition

$$\left(\frac{o - o_p}{o_r - o_p}\right)^2 + \left(\frac{s - s_p}{s_r - s_p}\right)^2 \geqslant 1, \tag{17}$$

where the maximum normal opening amount o_r depends on the tensile strength f_t and the energy release rate Gf_I of the mode I crack, and the tangential slipping amount s_r is related to the shear strength f_s and the energy release rate Gf_{II} of mode II crack.

3. Basic assumptions of seepage calculation

The rock mass is composed of rock blocks with pores and fractures, where fluid may exist in the fracture and the pores in the rock mass (i.e., the rock matrix). Likewise, the permeability of the rock mass is composed of two parts, the permeability of the rock matrix and the permeability of the fracture. The permeability of the fracture is mainly related to the aperture of the fracture, while the permeability of the rock matrix is mainly related to the porosity and the stress state.

As shown in Fig. 1, the tetrahedral element and the joint element form a unique topological connection in the 3D FDEM. The fluid can flow in the tetrahedral element and the broken joint element (i.e., fracture). The permeability of the rock matrix is represented by the fluid flow in the tetrahedral element, while the permeability of the fracture is characterized by fluid flow in the broken joint element. When a joint element breaks, the joint element becomes the channel of the fracture seepage.

Thus, the fully coupled three-dimensional model proposed in this paper is based on the following assumptions. (1) The permeability of the rock matrix is characterized by the porous seepage in the tetrahedral element according to Darcy's law. (2) The permeability of the fracture is represented by fracture seepage in the joint element based on the Cubic law. (3) Unbroken joint elements do not participate in the seepage calculation.

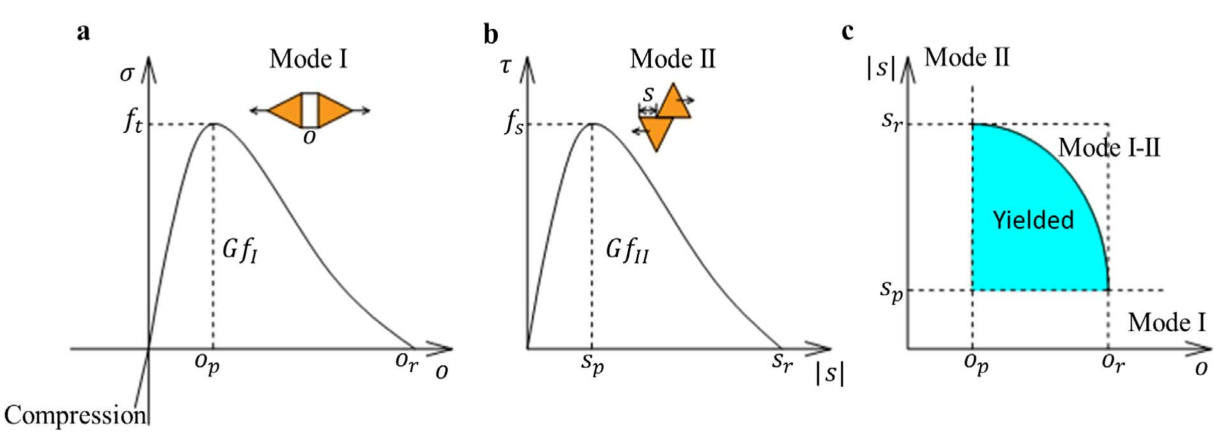


Fig. 3. Constitutive behavior of a joint element in 3D FDEM. (a) Relationship between normal bonding stress and normal opening amount in mode I; (b) Relationship between tangential bonding stress and tangential slipping amount in mode II; (c) Relationship between joint element yielded, failure and the normal opening amount and tangential slipping amount in mixed mode I-II [73].

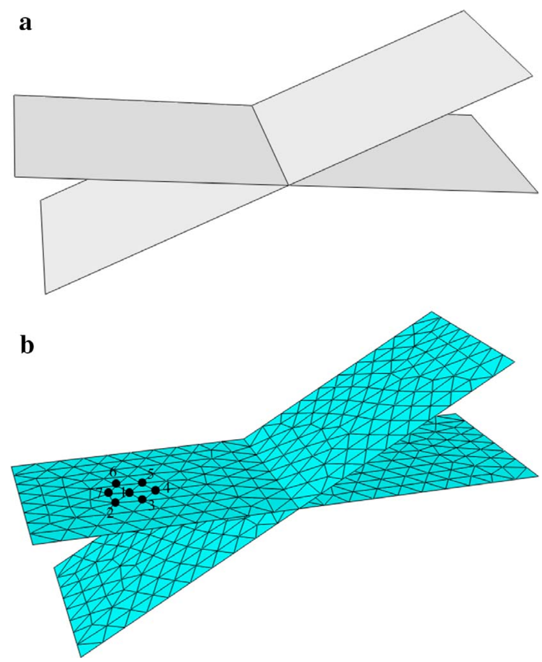


Fig. 4. Characterization of the fracture surface in the fully coupled 3D model. (a) Fracture surface; (b) Meshing of the fracture surface.

4. Fracture seepage and pore seepage calculation

4.1. Fracture seepage model

Since the joint element is inserted into the common face of the neighboring tetrahedral elements in 3D FDEM, the fracture surface is formed when the joint element breaks. Therefore, the fractures consist of a series of triangular faces, and fracture seepage only occurs on these triangular faces. Thus, fracture seepage in the fully coupled 3D model is in essence a two-dimensional seepage problem. As shown in Fig. 4a, there are two fracture surfaces, which are naturally transformed into a series of spatial triangular faces, as shown in Fig. 4b. Any one of the triangular faces constitutes a two-dimensional planar flow path. For example, the triangular face Δ123 formed by nodes 1, 2 and 3 constitutes a two-dimensional planar flow path. Next, we describe how the fluid pressure distribution over the entire fracture surface can be calculated based on the mesh as shown in Fig. 4b.

In order to facilitate a later discussion, we define the total head as

$$\phi = (P-\rho_w x_k g_k)/(\rho_w g), \tag{18}$$

where p is the fluid pressure, ρ_w is the fluid density, g_i ($i = 1, 2, 3$) is the component of gravitational acceleration, and g is the magnitude of the gravitational acceleration.

Taking node 1 in Fig. 4b as an example, there are six triangular faces that connect to node 1 (Δ123, Δ127, Δ176, Δ156, Δ145 and Δ134), where the triangular face Δ123 is taken as an example to describe the solution of the fluid flow in a triangular face. In the local coordinate system of the triangular face (x, y lies in the triangular face and the z axis is perpendicular to the triangular face), the coordinates of nodes 1, 2 and 3 are (x_k, y_k, 0) (where k is the node number, $k = 1, 2, 3$). The fluid pressures at nodes 1, 2 and 3 are assumed as p_1, p_2 and p_3, respectively. Assuming that the distribution of the fluid pressure in a triangular face obeys a linear distribution, then the total head in the triangular face is also subject to a linear distribution according to Eq. (18). Thus, the gradient of total head in the triangular face is constant and can be expressed as

$$\frac{\partial\phi}{\partial x_i} = \frac{1}{A}\int_A \frac{\partial\phi}{\partial x_i} dA. \tag{19}$$

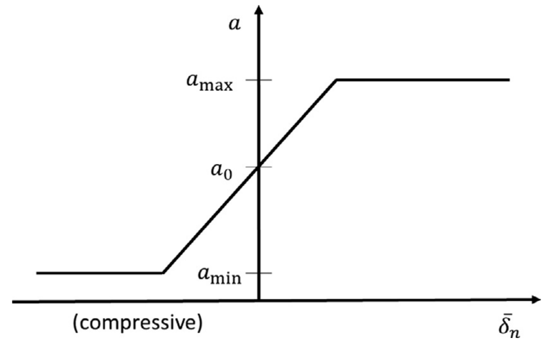


Fig. 5. Relationship between the hydraulic aperture and the average normal opening amount of a joint element.

According to the Gaussian divergence theorem, Eq. (19) can be written as

$$\frac{\partial\phi}{\partial x_i} = \frac{1}{A}\int_s \phi n_i ds = \frac{1}{A}\sum_{m=1}^{3} \bar{\phi}^m \in_{ij} \Delta x_j^m, \tag{20}$$

where A is the area of the triangular face, n_i is the outer normal unit vector, $\bar{\phi}^m$ is the average of the total head at edge m, Δx_j^m is the coordinate components difference of the two vertices of edge m, and $\in_{ij}$ is a two-dimensional permutation tensor $\in = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

According to the Cubic law, the flow rate along the i-direction can be expressed as

$$q_i = -\frac{a^3\rho_w g}{12\mu}\frac{\partial\phi}{\partial x_i}, \tag{21}$$

where q_i is the flow rate along the i-direction, a is the hydraulic aperture of the joint element, ρ_w is the fluid density, and μ is the fluid viscosity coefficient.

The hydraulic aperture a of the joint element is not constant, and its relationship to the average normal opening amount of the joint element is shown in Fig. 5.

In this way, the hydraulic aperture of the joint element can be expressed as

$$a = a_0 + \bar{\delta}_n, \tag{22}$$

where a_0 is the initial hydraulic aperture of the joint element, and $\bar{\delta}_n$ is the average normal opening amount of the joint element (positive denotes opening). $\bar{\delta}_n$ is equal to the average value of the opening amount of the three nodes, i.e., $\bar{\delta}_n = (\delta_{n1} + \delta_{n2} + \delta_{n3})/3$, where δ_{n1}, δ_{n2} and δ_{n3} are the normal opening amounts at the three nodes on the top and bottom faces respectively, as shown in Fig. 6. Since the hydraulic aperture cannot have a value less than or equal to zero in the seepage calculation, a minimum $a_{\min}$ is set. If the hydraulic aperture calculated by Eq. (22) is less than the minimum, the hydraulic aperture is set to $a_{\min}$. To describe the permeability decrease as the joint element closes, $a_{\min}$ should be set to a small value. According to Eq. (22), the effect of the minimum aperture on the flow rate is shown in Fig. 7, where q_0 is the flow rate at $a = a_0$. The minimum of q/q_0 decreases with the decrease of the minimum aperture $a_{\min}$. If $a_{\min} = 0.1\ a_0$, The minimum of q/q_0 is 0.001, which means that the calculation error of the flow rate is less than the thousandth of q_0. Thus, $a_{\min}$ is set 0.1 a_0 in this paper,

Thus, substituting Eq. (20) into Eq. (21), the flow rate along the i-direction can be obtained as

$$q_i = -\frac{a^3\rho_w g}{12\mu}\left(\frac{1}{A}\sum_{m=1}^{3} \bar{\phi}^m \in_{ij} \Delta x_j^m\right). \tag{23}$$

Thus, the flow into node 1 per unit time can be calculated by

$$Q^n = -\frac{q_i n_i^{(n)} S^{(n)}}{3}, \tag{24}$$

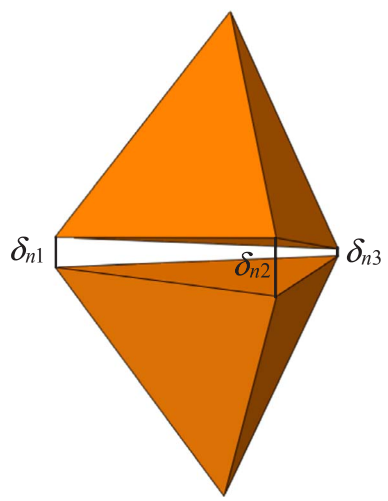


Fig. 6. Normal opening amount of a joint element.

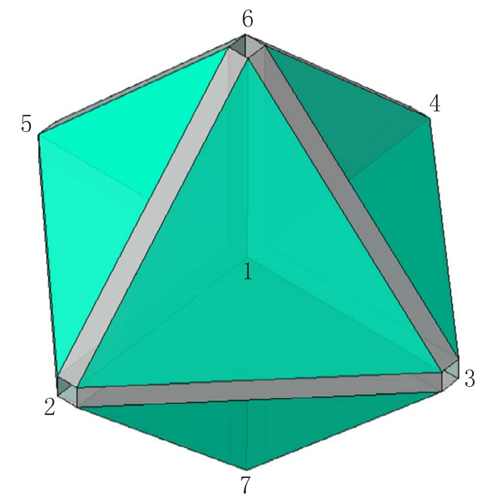


Fig. 8. 3D porous seepage mode in the fully coupled 3D model.

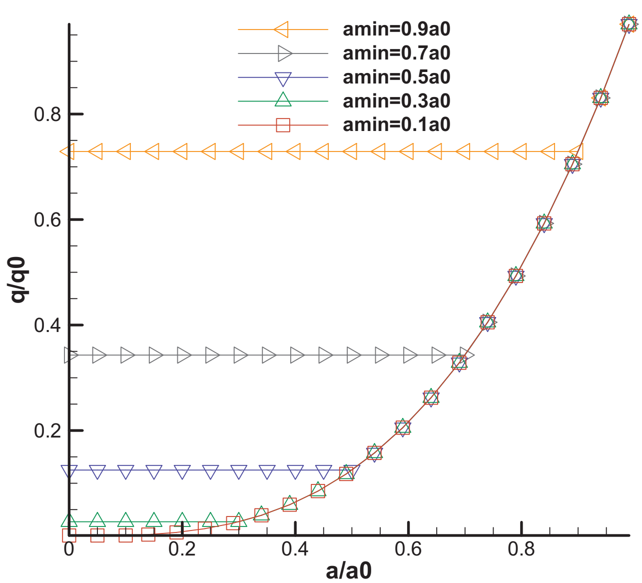


Fig. 7. The sensitivity analysis of flow rate to the minimum aperture $a_{\min}$.

where $n_i^{(n)}$ is the outer normal unit vector of the face opposite node n .

Thus, we obtain the flow $Q_{\Delta 123}$ into node 1 from the triangular face $\Delta 123$ per unit time. Similarly, we can obtain the flow into node 1 from other triangular faces directly connecting to the node. Thus, the total flow into node 1 per unit time can be expressed as

$$Q_{\text{total}} = Q_{\Delta 123} + Q_{\Delta 127} + Q_{\Delta 176} + Q_{\Delta 156} + Q_{\Delta 145} + Q_{\Delta 134}. \tag{25}$$

Finally, the fluid pressure of node 1 at the current step can be updated as follows [75]

$$p_{t+\Delta t}^{f} = p_t^{f} + K_w Q\frac{\Delta t}{V} - K_w\frac{\Delta V}{V_m}, \tag{26}$$

where p_t^f is fluid pressure of node 1 at the previous time step, K_w is the fluid bulk modulus, Δt is the time step, $\Delta V = V - V_0$, and $V_m = (V + V_0)/2$, where V and V_0 are the nodal volume at the previous and current time step, respectively.

4.2. Three-dimensional porous seepage model

According to Darcy's law, the flow rate of per unit sectional area along the *i*-direction can be expressed as

$$q_i = -\frac{k_{ij}}{\mu}\frac{\partial(p - \rho_w g_k x_k)}{\partial x_j}, \tag{27}$$

where g_i (i = 1, 2, 3) is the component of the acceleration of gravity, k_{ij} is the intrinsic permeability tensor, μ is the fluid viscous coefficient, and p is the pore pressure.

According to the definition of Eq. (18), Eq. (27) can be rewritten as

$$q_i = -\rho_w g\frac{k_{ij}}{\mu}\frac{\partial\phi}{\partial x_j}. \tag{28}$$

For any given volume V, the change in pore pressure can be written as [76]

$$\frac{\partial p}{\partial t} = \frac{M}{V}\left(Q_{total} - \alpha\frac{\partial V}{\partial t}\right), \tag{29}$$

where Q_{total} is the net flow into volume V per unit time, M is the Biot modulus, and α is the Biot coefficient.

Because of the special topological connection between the joint element and the tetrahedral element in 3D FDEM, we construct the 3D porous seepage model as shown in Fig. 8, where the numbered nodes are the shared nodes of the tetrahedral elements and the joint elements. In this model, we assign 1/4 of the mass of all tetrahedral elements connecting to node 1 equal to the node, and the pore pressure of the node represents the pore pressure in the zone around the node. In the following, we describe how the pore pressure distribution of the entire continuum can be calculated based on the topological connections shown in Fig. 8.

Taking node 1 in Fig. 8 as an example, there are 8 tetrahedral elements connected to node 1 (T1236, T1346, T1456, T1526, T1237, T1347, T1457 and T1527). Since the pore pressure of nodes 2, 3, 4, 5, 6, or 7 may be different from the pore pressure of node 1, porous seepage may occur in these tetrahedral elements. Take the tetrahedral element T1236 connected to node 1 as an example. Assuming the pore pressure in the tetrahedral element obeys a linear distribution, the total head in the tetrahedral element also follows a linear distribution. Thus, the gradient of the total head in the tetrahedral element is constant and can be expressed as

$$\frac{\partial\phi}{\partial x_i} = \frac{1}{V}\int_A \frac{\partial\phi}{\partial x_i}dV. \tag{30}$$

According to the Gaussian divergence theorem, Eq. (30) can be written as

$$\frac{\partial\phi}{\partial x_i} = \frac{1}{V}\int_s \phi n_i dV = -\frac{1}{3V}\sum_{l=1}^{4}\phi^l n_i^{(l)} S^{(l)}, \tag{31}$$

where V is the volume of the tetrahedral element, n_i is the outer normal unit vector of the l face of the tetrahedral element, ϕ^l is the total head of the vertex opposite to the l face, and $S^{(l)}$ is the area of the l face.

Next, substituting Eq. (31) into Eq. (28), the flow rate along the i direction can be obtained by

$$q_i = \rho_w g \frac{k_{ij}}{\mu}\left(\frac{1}{3V}\sum_{l=1}^{4}\phi^l n_j^{(l)} S^{(l)}\right). \tag{32}$$

It should be noted that according to Eq. (32), the flow rate in a tetrahedral element is not zero under the effect of gravity even if the pore pressure of the four nodes of the tetrahedral element are all zero, which is unreasonable. The flow rate between the nodes should decrease as the saturation decreases. If the saturation is zero, the flow rate should also be zero. The fluid will not flow out from a zone where the saturation is zero. Thus, Eq. (32) should be multiplied by a coefficient f_s, which is a function of the average saturation

$$f_s = \bar{s}^2(3-2\bar{s}), \tag{33}$$

where $\bar{s}$ is the average saturation of the outflow node. The empirical function has the following properties: $f_s = 0$ if $\bar{s} = 0$ and $f_s = 1$ if $\bar{s} = 1$. That is, if the average saturation is zero, the fluid flow in the tetrahedral element is zero; if the node is fully saturated, the flow rate is not reduced. Moreover, the derivatives of the function at $\bar{s} = 0,1$ are both zero, which is reasonable physically. Therefore, the final flow rate along the i-direction can be expressed as

$$q_i = -\rho_w g \frac{k_{ij}}{\mu}\frac{\partial \phi}{\partial x_j} f_s. \tag{34}$$

Thus, the flow rate into node 1 per unit time can be calculated by the following equation

$$Q^n = -\frac{q_i n_i^{(n)} S^{(n)}}{3}, \tag{35}$$

where $n_i^{(n)}$ is the outer normal unit vector of the face opposite node 1.

Thus, we obtain the flow Q_{T1236} into node 1 from tetrahedral element T1236. Similarly, we can obtain flow into node 1 from other tetrahedral elements directly connecting to node 1. In this way, the total flow into node 1 can be expressed as

$$Q_{total} = Q_{T1236} + Q_{T1346} + Q_{T1456} + Q_{T1526} + Q_{T1237} + Q_{T1347} + Q_{T1457} + Q_{T1527}.$$

Then, according to the differential form of Eq. (29), the pore pressure of node 1 at the next time step is given by

$$p_{t+\Delta t}^p = p_t^p + \frac{M(Q_{total}-\alpha\Delta V_{mech})}{V}\Delta t. \tag{36}$$

Thus, the pore pressure of node 1 at the next time step is updated. Similarly, the pore pressure of the other nodes can be obtained according to the above method. From this, we can obtain the evolution of the pore pressure field of the entire domain.

If the node pressure calculated by Eq. (36) is negative, then the node pressure is set to zero, which means that the fluid equivalent to node 1 is not sufficient to fill the equivalent volume of the node and the saturation of the node will decrease. The saturation of the node is updated according to the following equation

$$s = s_0 + \frac{Q_{total}\Delta t-\Delta V_{mech}}{nV}, \tag{37}$$

where n is porosity of rock matrix, and s_0 is the saturation of node 1 at the previous step. The fluid pressure of the node will be zero as long as the saturation of the node at the current time step is less than one. If saturation $s > 1$, the saturation is set to one. If the fluid pressure calculated by Eq. (36) is greater than zero, the change in nodal volume alters the fluid pressure of the node. Conversely, if the fluid pressure calculated by Eq. (36) is less than 0, the variation of nodal volume changes the saturation of the node.

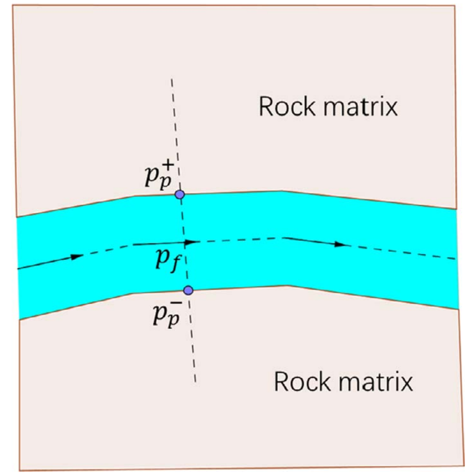


Fig. 9. Coupling treatment of porous seepage and fracture seepage.

The time step needs to be less than the critical time to ensure numerical stability because the porous seepage calculation adopts an explicit algorithm. The critical time step is given by

$$\Delta_f = \min\left[\frac{V}{M\sum_i k_i/\mu}\right], \tag{38}$$

where k_i is the isotropic intrinsic permeability.

4.3. Coordinated treatment of fracture seepage and porous seepage

Because both porous seepage and fracture seepage are taken into account, the fluid exchange between porous seepage and fracture seepage exists in the fracture, as shown in Fig. 9. Thus, how to deal with the fluid flow between the fractures and rock matrix is the key to solve the pressure distribution within the entire domain. We address the problem using the assumption that the pore pressure and the fracture fluid pressure satisfy the continuity equation at the fractures [74]

$$p_+^p = p^f = p_-^p, \tag{39}$$

where p^f is the fluid pressure in the fracture, and p_+^p and p_-^p are the pore pressure on the upper and lower walls of the fracture, respectively. For the application of the continuity equation Eq. (39), we use the following method. That is, after the fracture seepage calculation, the fracture pressure at the boundary of the fracture is taken as the boundary condition of the subsequent porous seepage; then the pore seepage calculation is carried out and the pore pressure at the fracture boundary is taken as the initial value of fracture pressure in seepage calculation at the next time step. The repetition of the above process achieves the application of the continuity equation Eq. (39).

5. Hydro-mechanical coupling

5.1. The basic concept of hydro-mechanical coupling

In Section 4, we discuss the calculation of fracture seepage and porous seepage, respectively. In this section, we discuss the basic idea of the 3D coupled hydro-mechanical model from the global perspective, and give the flow chart of the coupled model. In the 3D fully coupled model, the hydro-mechanical coupling mixes porous seepage-stress coupling and fracture seepage-stress coupling. Hydro-mechanical coupling is built on the following processes. (1) The fluid flow in the fracture (i.e. broken joint element) will exert a corresponding fluid pressure. The fluid pressure make the fractures open or close, which

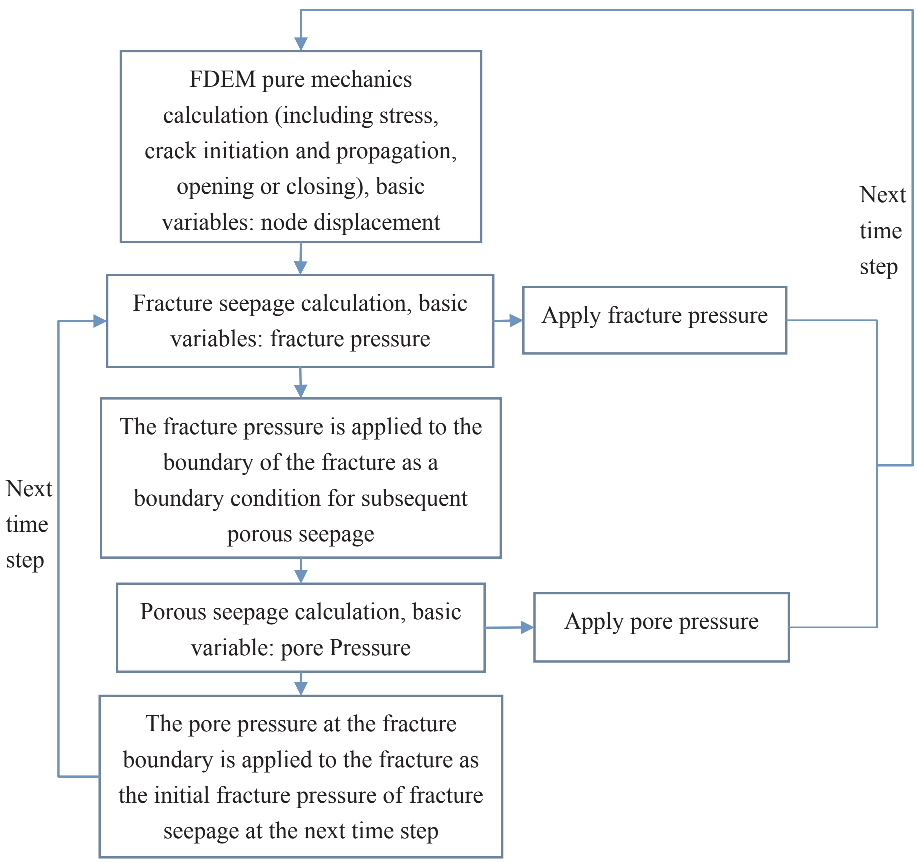


Fig. 10. The flow chart of fully coupled 3D hydro-mechanical model.

will conversely affect the fluid flow in the fracture. (2) The fluid in the rock matrix will apply pore pressure and make the rock matrix deform, which will conversely affect the fluid flow and pore pressure in the rock matrix. The calculation process of the fully coupled 3D hydro-mechanical model is shown in Fig. 10. It can be seen from the flow chart that the pure mechanical calculation is carried out first. Then, the stress field and the crack aperture are fixed, and the calculation of fracture seepage and porous seepage is carried out. Finally, the fracture pressure and pore pressure is taken as the external load of pure mechanical calculation at the next time step. By repeating the above process, the hydro-mechanical coupling is achieved. The entire hydro-mechanical coupling calculation is carried out using the staggered scheme. Since it is an explicit calculation, no iteration is required. However, the time step should be less than the critical time step to ensure that the calculation results converge.

5.2. The effect of fracture seepage on the stress field

Fluid flow in a fracture applies normal fluid pressure on the fracture wall. As shown in Fig. 11, the fluid pressure acting on the tetrahedral elements connected by a broken joint element is given by

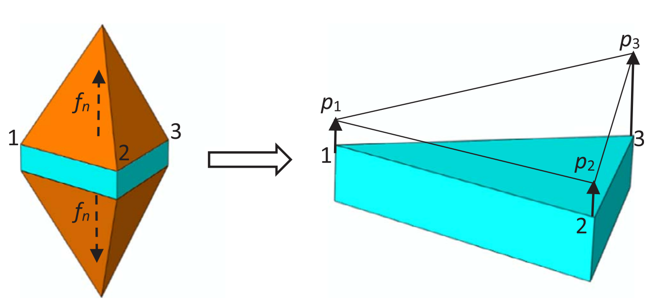


Fig. 11. The distribution of the fluid pressure.

$$f_p = \frac{p_1 + p_2 + p_3}{3}. \tag{40}$$

where p_1, p_2 and p_3 are fluid pressure at nodes 1, 2 and 3, respectively.

Thus, the fluid pressure acting on the tetrahedral elements is obtained. The fluid pressure is then applied to the triangular faces of the two tetrahedral elements connected by the joint element as a linear distributed load. Like the contact force and other external loads, the fluid pressure is also distributed to the nodes of the tetrahedral elements.

5.3. The effect of stress changes and crack generation (joint element breaking) on the fracture seepage

The change in the stress affects the aperture of the fracture and then changes the permeability of the fracture. When the aperture of the fracture changes, according to Eq. (21), the flow rate will change and the fluid pressure in the fracture will also change.

The breaking of the joint element causes the network of the fracture seepage to change. Then, the parameters such as flow rate and fluid pressure in the fracture will also change, i.e., the fracture seepage field is changed. Through the breaking of the joint element and Eq. (21), the effect of crack generation on fracture seepage is handled automatically in the fully coupled 3D model. The traditional coupled seepage-stress model without considering crack generation cannot address this effect.

5.4. The effect of porous seepage on the stress field

The porous seepage in the rock matrix will exert the effect of the body force on the rock matrix. According to the effective stress principle, the effective stress in the rock matrix can be expressed as

$$\sigma'_{ij} = \sigma_{ij} - \alpha p \delta_{ij}. \tag{41}$$

Therefore, the pore pressure causes the stress field in the rock

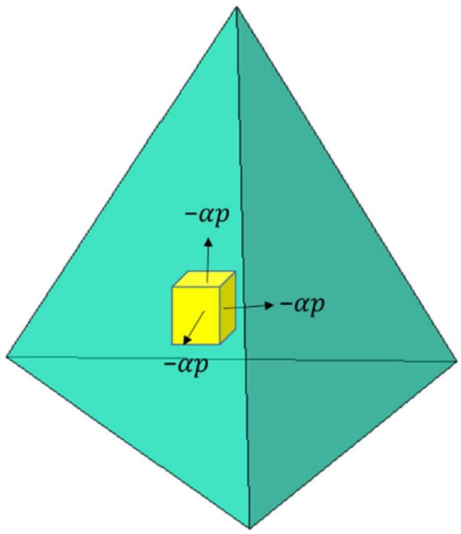


Fig. 12. The effect of pore pressure on the rock matrix.

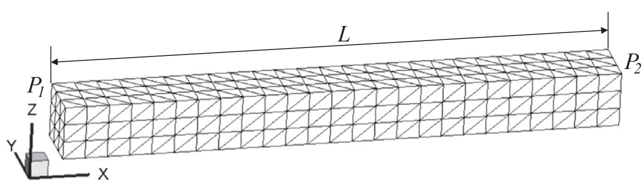
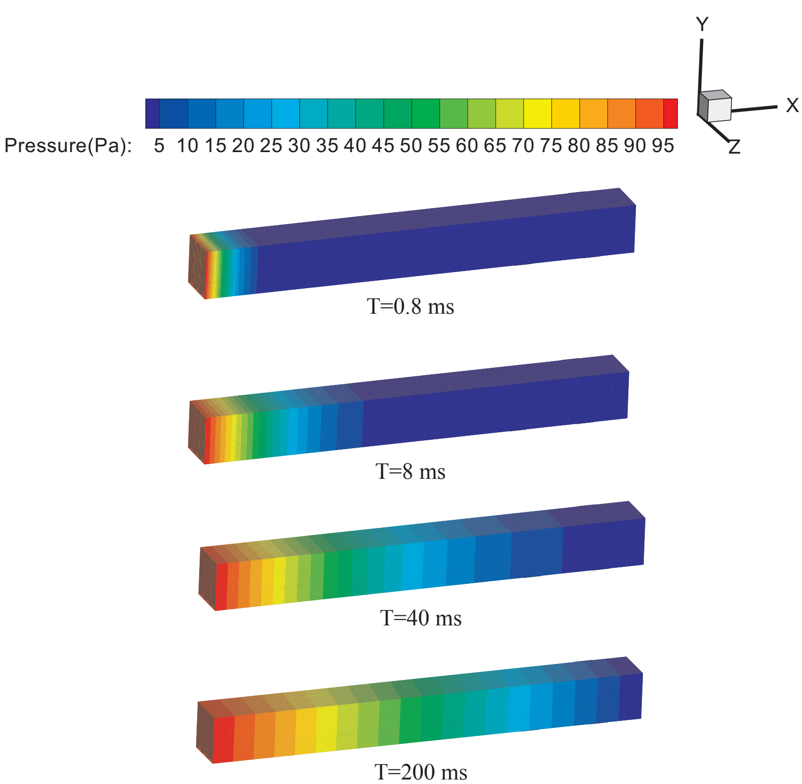


Fig. 13. The calculated mesh.

matrix to change. As shown in Fig. 12, the stress increment due to the pore pressure is given by

$$\Delta\sigma_{ij} = -\alpha p\delta_{ij}. \tag{42}$$

In this way, we obtain the stress increment caused by the pore pressure, which will apply to the tetrahedral element as a body load.



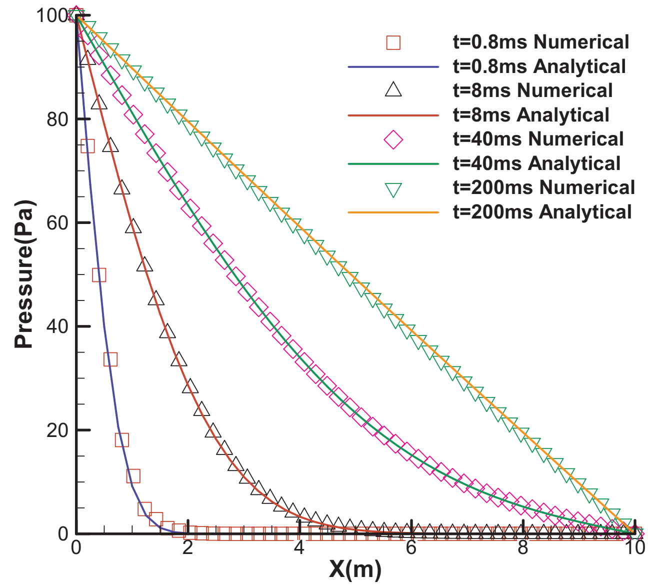


Fig. 15. Numerical and analytical solutions of pore pressure at different times.

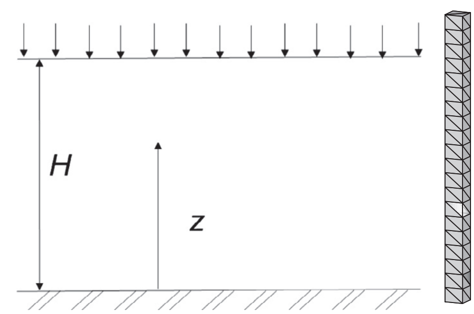


Fig. 16. The consolidation model and computational mesh.

Fig. 14. Distribution of pore pressure at different times.

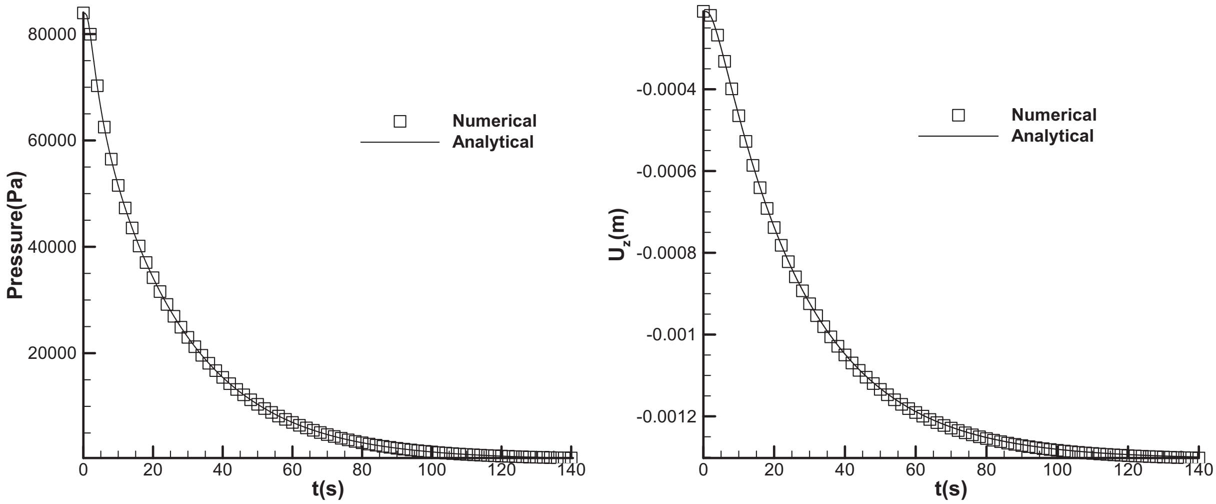


Fig. 17. The variation of pore pressure and vertical displacement of the midpoint with time.

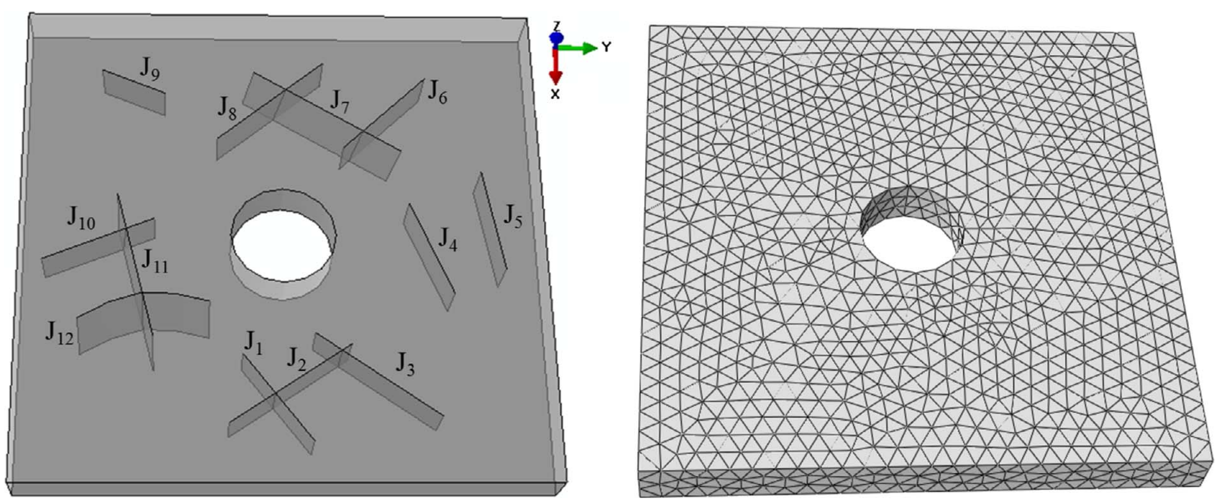


Fig. 18. Geometric model and the calculated mesh.

Table 1
Mechanical input parameters for hydraulic fracturing.

Parameters	Value
Rock	
Bulk density, ρ (kg/m^3)	2300
Young's modulus, E (GPa)	18
Poisson's ratio, v	0.18
Joint element	
Tensile strength, f_t (MPa)	15
Internal cohesion, c (MPa)	25
Friction angle of intact material, ϕ (°)	30
Friction angle of fractures, ϕ (°)	30
Fracture energy release rate, Gf_I (J/m^2)	10.5
Fracture energy release rate, Gf_{II} (J/m^2)	105
Normal contact penalty, p_n (GPa)	1000
Tangential contact penalty, p_t (GPa/m)	1000
Fracture penalty, p_f (GPa)	1000

5.5. The effect of stress variation on porous seepage

The compressibility of the rock has two effects on the seepage process. (1) The pore pressure and the stress field change will change the pore size in the rock matrix, i.e., the porosity is a function of the stress state and pore pressure. (2) The change in the pore size causes the variety in permeability.

We use the following coupling equation to characterize the effect of the stress field and pore pressure on the intrinsic permeability [32]

$$k(\sigma,p) = \xi k_0 e^{-\beta(\sigma_{ii}/3-\alpha P)}$$
$$\sigma_{ii} = \sigma_{11} + \sigma_{22} + \sigma_{33}, \tag{43}$$

where ξ is the mutation coefficient, which is used to account for the permeability increasing once the element reaches the damage state. Since the tetrahedral element is considered to be linearly elastic in the fully coupled 3D model, the coefficient is set to 1. $\sigma_{ii}/3$ is the average total stress, and β is the coupling parameter, which characterizes the effect of stress on the permeability. The coefficient can be determined by experimental test. α is the Biot coefficient, and p is the pore pressure.

5.6. Fracture generation and propagation

In 3D FDEM, joint elements are inserted between neighboring tetrahedral elements. Under the action of the fluid pressure and other loads, some joint elements may break, i.e., new fractures are generated. Thus, crack initiation and propagation in the fully coupled model is achieved automatically. The newly generated fracture constitutes the network channel of the fracture seepage with the pre-existing fracture.

6. Examples

6.1. Example 1: Unsteady seepage

As shown in Fig. 13, a fully saturated model has a length of 10 m and a width of 1 m. The porosity of the model n is 0.1, and the intrinsic permeability k is 1×10^{-11} m^2. The fluid pressure is fixed with $P_1 = 100$ kPa at the left side of the model and $P_2 = 0$ kPa at the right side. The fluid density ρ_w is 1000 kg/m^3, the fluid viscosity coefficient μ is 0.001 Pa s, and the fluid bulk modulus K_w is 2.2 GPa. The variation of pore pressure over time in the model is studied. The calculated mesh is shown in Fig. 13 with a total of 1835 tetrahedral elements and 3988 joint elements. A constant integration time step of 2×10^{-7} s is used.

The analytical solution of this problem is given by

$$\boxed{P(x,t) = P_1 + \frac{x}{L}(P_2 - P_1) + \frac{2}{\pi}\sum_{n=1}^{\infty} e^{-\kappa n^2 \pi^2 t/L^2}\left(\frac{P_2\cos(n\pi) - P_1}{n}\right)\sin\frac{n\pi x}{L},} \tag{44}$$

where P_1 is the fluid pressure at the left side, P_2 is the fluid pressure at the right side, L is the length of the model, t is the time, x is the distance from the left side, and $\kappa = (k/\mu)M$, where k is the intrinsic permeability, μ is the viscous coefficient, and M is the Biot modulus of porous

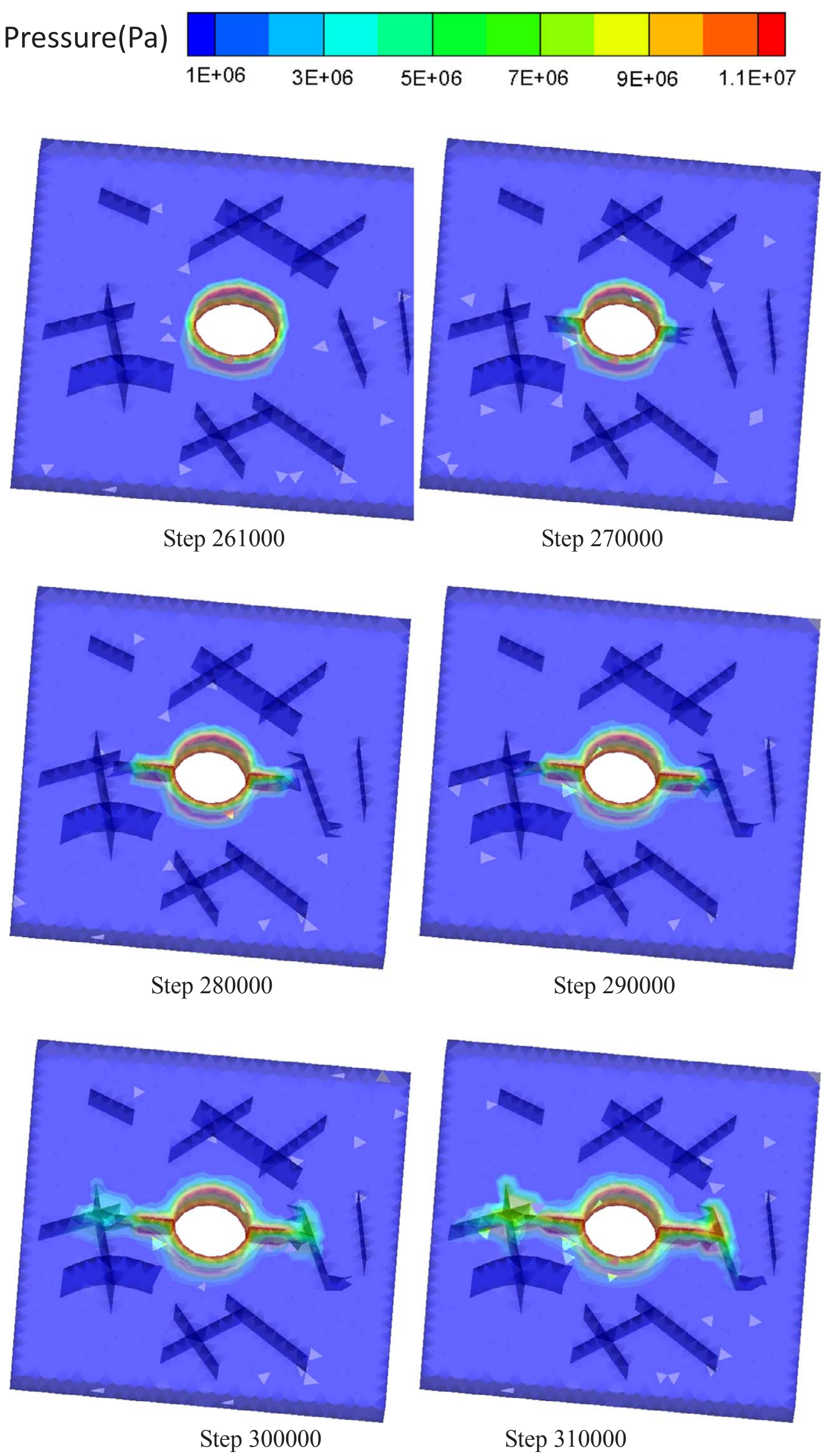


Fig. 19. The distribution of fluid pressure and crack propagation during hydraulic fracturing.

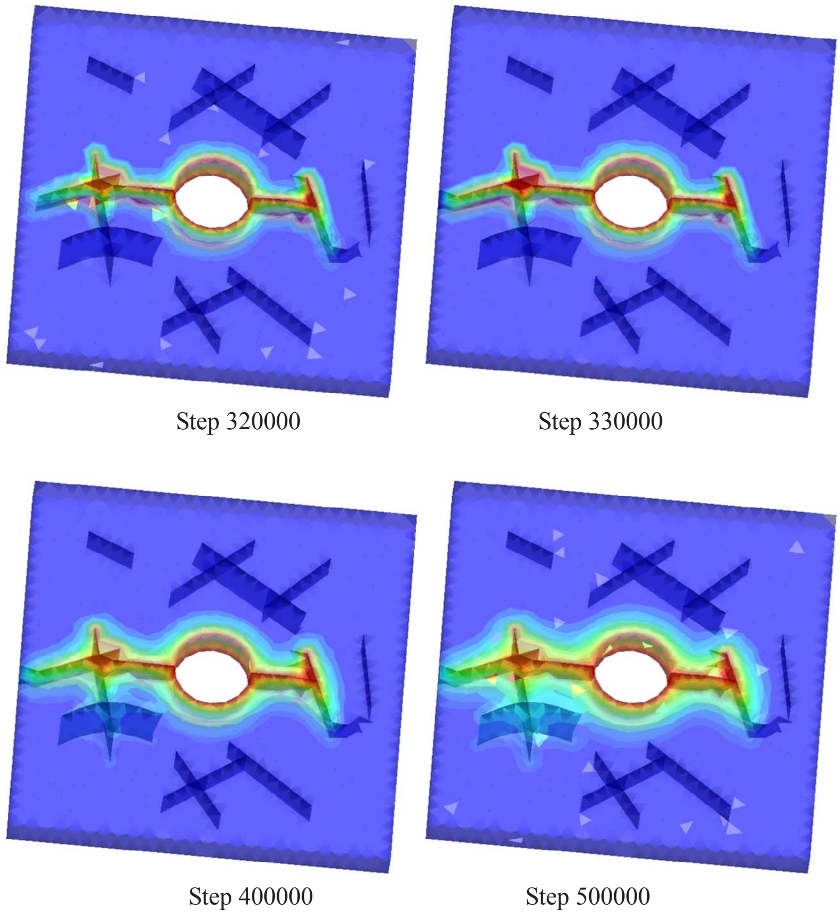


Fig. 19. (*continued*)

media.

The pore pressure distribution at $t = 0.8$ ms, 8 ms, 40 ms, and 200 ms is shown in Figs. 14 and 15. It can be seen that the numerical solutions agree well with the analytical solutions. The results show that this fully coupled 3D model is correctly calculating unsteady-state seepage. It can avoid the flow lag phenomenon that exists in the equivalent approach [34,68,69] when dealing with unsteady porous seepage. Since steady-state seepage is the final result of unsteady-state seepage, the model also correctly addresses steady-state seepage. As shown in Fig. 15, the pore pressure in the model is linearly distributed at $t = 200$ ms (i.e., the time at which seepage reaches steady state), which is consistent with the pressure distribution at steady state.

6.2. Example 2: Unsteady porous seepage-stress coupling

As shown in Fig. 16, a saturated semi-infinite soil layer with a thickness of 20 m is laid on a rigid impermeable base. The soil layer is assumed to be a uniform elastomer, and the fluid flow in it is described using isotropic Darcy's law. Under the undrained condition, a constant surface load $p_z = 1 \times 10^5$ Pa is applied to the top of the soil layer [76].

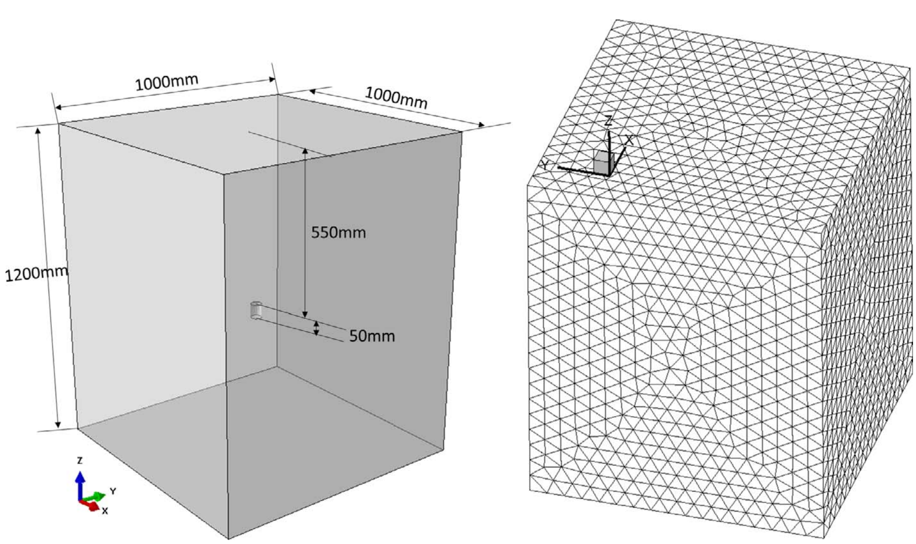


Fig. 20. The specimen size and calculation mesh.

Table 2
In situ stress.

Case	σ_x (MPa)	σ_y (MPa)	σ_z (MPa)
I	4.0	4.0	1.0
II	1.0	3.0	5.0
III	1.0	1.0	1.0

Table 3
Mechanical input parameters.

Parameters	Value
Solid	
Bulk density, ρ (kg/m^3)	2300
Young's modulus, E (GPa)	6
Poisson's ratio, ν	0.25
Joint element	
Tensile strength, f_t (MPa)	3
Internal cohesion, c (MPa)	15
Friction angle of intact material, ϕ (°)	30
Friction angle of fractures, ϕ (°)	30
Fracture energy release rate, Gf_I (J/m^2)	10.5
Fracture energy release rate, Gf_{II} (J/m^2)	105
Normal contact penalty, p_n (GPa)	300
Tangential contact penalty, p_t (GPa/m)	300
Fracture penalty, p_f (GPa)	300

Initially, the applied surface loads are all borne by the fluid, but they are continually transferred into the soil as the fluid is gradually drained from the soil surface. The analytical solution of this consolidation problem can be obtained according to Biot's consolidation theory [77].

In this problem, the analytical solution of pore pressure is given by

$$p = 2p_0 \sum_{m=0}^{\infty} \frac{\sin(a_m \hat{z})}{a_m} e^{-a_m^2 \hat{t}}, \tag{45}$$

where $p_0 = \frac{\alpha}{\alpha_1}\frac{1}{S}p_z$, $a_m = \frac{\pi}{2}(2m+1)$; $\hat{z} = \frac{H-z}{H}$ and $\hat{t} = \frac{ct}{H^2}$, where $c = k/(\mu S)$, $S = \frac{1}{M} + \frac{\alpha^2}{\alpha_1}$ is the storage coefficient, $\alpha_1 = K + \frac{4}{3}G$, M is the Biot modulus, and α is the Biot coefficient.

The analytical solution for the vertical displacement is given by

$$u_z = \frac{Hp_z}{\alpha_1}\left(2\alpha\frac{p_0}{p_z}\left[\sum_{m=0}^{\infty} \frac{\cos(a_m \hat{z})}{a_m^2} e^{-a_m^2 \hat{t}}\right] + \hat{z} - 1\right). \tag{46}$$

The calculated parameters are as follows: the bulk modulus of the porous media K is 5×10^8 Pa, the shear modulus G is 2×10^8 Pa, the Biot modulus M is 4×10^9 Pa, the Biot coefficient α is 1.0, the intrinsic permeability k is 1×10^{-11} m^2, the fluid density ρ_w is 1000 kg/m^3, and the viscosity coefficient μ is 0.001 Pa s. The calculation mesh is as shown at the right of Fig. 16 with a total of 226 tetrahedral elements and 534 joint elements. A constant integration time step of 2×10^{-6} s is used. The side faces and the bottom of the model are fixed along the normal direction. At $z = H/2$, we set up a point to monitor the variation of pore pressure and displacement with time. As shown in Fig. 17, the numerical solution agrees well with the analytical solution, which verifies the correctness of the fully coupled 3D model in addressing unsteady seepage-stress coupling.

6.3. Hydraulic fracturing application example

As shown in Fig. 18(a), there is a rectangular sample with an edge length of 10 m and a thickness of 1 m. An injection hole with a radius of 1 m is in the center of the rectangular sample. In addition, there are a total of 12 pre-existing fractures in the sample. These pre-existing fractures are numbered as J1-J12, where J7 is an inclined plane that is not perpendicular to the xy plane, and J12 is an irregular surface, also not perpendicular to the xy plane. This problem is a true three-dimensional problem and cannot be considered a plane stress or a plane strain problem. The in situ compressive stresses are $\sigma_{xx} = 5$ MPa, $\sigma_{yy} = 20$ MPa, and $\sigma_{zz} = 10$ MPa. The fluid pressure in the injection hole is 12 MPa. The problem is simulated by the fully coupled 3D model for studying the interaction between the hydraulic fracture and the pre-existing fracture. The pure mechanics input parameters are listed in Table 1, while the fluid parameters are as follows: the fluid density $\rho_w = 1000$ kg/m^3, the viscosity coefficient of the fluid $\mu = 0.001$ Pa s, the fluid bulk modulus $K_w = 2.2$ GPa, the porosity of the rock matrix $n = 0.1$, the intrinsic permeability $k = 5 \times 10^{-11}$ m^2, and the initial aperture of the pre-existing fractures $a_0 = 5 \times 10^{-4}$ m. The calculated mesh is shown in Fig. 18(b) with a total of 11608 tetrahedral elements and 25176 joint elements. A constant integration time step of 5×10^{-7} s is used. The results obtained from this fully coupled 3D model are shown in Fig. 19.

As shown in Fig. 19, the hydraulic fractures initiate from the boundary of the injection hole along the direction of the maximum principal stress of the in situ stress (i.e., along the y-axis) as shown in steps 270000–290000 in Fig. 19. The hydraulic fractures intersect with the J11 and J4 fractures, and the fluid infiltrates into these pre-existing fractures preferentially after they intersect with the hydraulic fractures, as shown in steps 300000–320000 in Fig. 19. However, the hydraulic fractures do not cross the J11 and J4 fractures but initiate from the end of the pre-existing fracture and deflect to the direction of the maximum principal stress (the y-axis). For example, at the left side, the hydraulic fracture initiates from the end of fracture J10 and extends along the y-axis direction, while at the right side, the hydraulic fracture initiates along the end of fracture J4 and continues to extend along the direction of the y-axis. This is due to the control effect of the in situ stress; the hydraulic fracture does not extend along the direction of the original pre-existing fractures but along the direction of the y-axis (i.e., the direction of the maximum principal stress), as shown in steps 330000–500000.

In addition, from the evolution of the fluid pressure in Fig. 19, it can be seen that the fluid preferentially infiltrates along the hydraulic fractures since the permeability of the rock matrix is very small relative to the newly created fractures and the pre-existing fractures, as shown in step 280000 in Fig. 19. When the hydraulic fracture intersects with the pre-existing fracture, the fluid preferentially infiltrates into the pre-existing fractures that directly connect to the hydraulic fractures, as shown in steps 270000–330000 in Fig. 19. Thus, the fluid pressure displays a strip distribution along the connected fracture network formed by the hydraulic fracture and the pre-existing fractures that connect to the hydraulic fracture directly. As the time steps increase, the fluid gradually infiltrates the surrounding rock matrix from the hydraulic fractures and the pre-existing fractures, and the fluid pressure is zonally distributed along the connected fracture network, as shown step 500000 in Fig. 19.

It can be seen from Fig. 19 that the proposed fully coupled 3D model can capture crack initiation and propagation, the interaction of hydraulic fractures and pre-existing fractures, and the evolution of fluid pressure during the fracturing process with three-dimensional arbitrary complex fracture networks. In contrast, the conventional method is often difficult to perform three-dimensional fracturing simulation with complex fracture networks.

6.4. Hydraulic fracturing under different in situ stress

In situ stress is the main factor controlling the direction of hydraulic fracture propagation [32]. In this part, we use the three-dimensional coupled hydro-mechanical model to study the effect of in situ stress on hydraulic fracturing. As shown in Fig. 20(a), it is a specimen with an edge length of 1000 mm, a height of 1200 mm, and an injection hole with a diameter of 40 mm in it. We study the hydraulic fracture propagation behavior of the specimen under different in situ stress

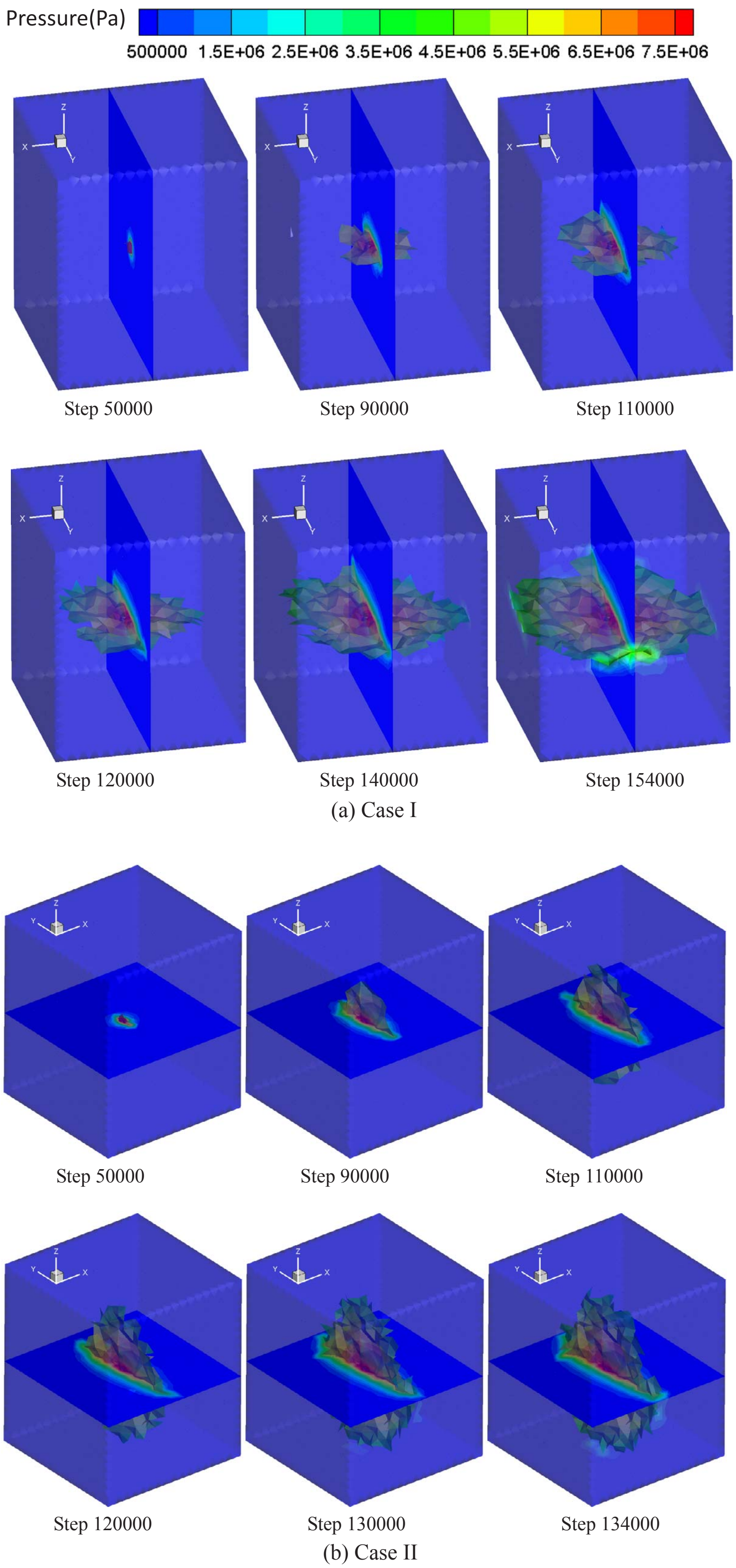


Fig. 21. Crack propagation behavior and fluid pressure evolution under different in situ stress conditions.

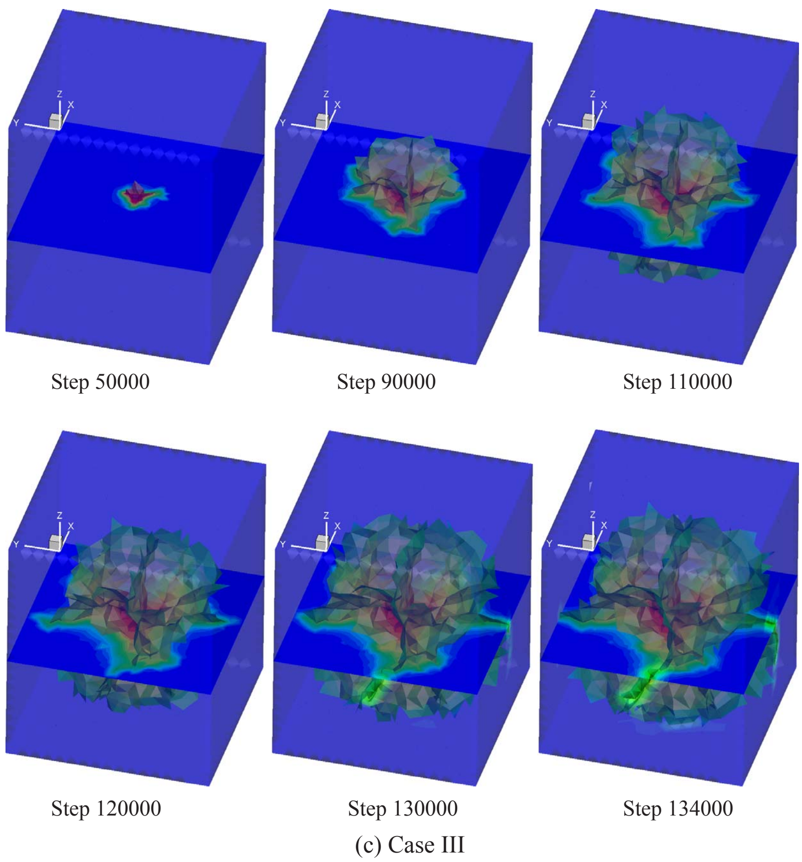


Fig. 21. (*continued*)

conditions (as shown in Table 2). The calculation mesh is shown in Fig. 20(b) with a total of 35858 tetrahedral elements and 73965 joint elements. Fluid pressure is applied to the boundary of the internal injection hole, the initial pressure is 3MPa. The increasing rate of fluid pressure in the injection hole is 100 Pa/step and keeps constant when increasing to 8.1 MPa. The mechanical parameters used are shown in Table 3.While the fluid parameters are as follows: the fluid density $\rho_w = 1000\ \mathrm{kg/m^3}$, the viscosity coefficient of the fluid $\mu = 0.001$ Pa s, the fluid bulk modulus $K_w = 2.2$ GPa, the porosity of the solid matrix $n = 0.1$, the intrinsic permeability $k = 1 \times 10^{-12}\ \mathrm{m^2}$, and the initial aperture $a_0 = 5 \times 10^{-5}$ m. A constant integration time step of 2.5×10^{-9} s is used.

As shown in Fig. 21, it is the fracture propagation process and the evolution of fluid pressure in the specimen under three different in situ stress conditions. As shown in Fig. 21(a) and (b), the hydraulic fracture are substantially perpendicular to the direction of the minimum principal stress in the in situ stress. In case I and II, the hydraulic fracture is very consistent with the experimental observation, as shown in Fig. 22(a) and (b). In Fig. 21(c), since the three components of the in situ stress are the same magnitude, the propagation direction of the hydraulic fracture is uncertain. Thus, the hydraulic fractures are very irregular and with many bifurcations. In addition, the distribution of fluid pressure in the cross section of Fig. 21 shows that there is a distribution of fluid pressure in the solid matrix around the fracture except for the distribution of fluid pressure at the fracture, which also means that the fluid penetrates into the solid matrix.

7. Conclusions

This paper presents a fully coupled 3D model for simulating hydraulic fracturing. In this model, the fluid flow in fractures is represented by fracture seepage, while the fluid flow in the rock matrix is characterized by the real porous seepage. The fully coupled 3D model is verified in detail, and the numerical and analytical solutions are in good agreement. Combined with the mechanical calculation of FDEM, the fully coupled 3D model can simulate 3D hydraulic fracturing with arbitrary complex fracture networks. It can capture crack initiation and propagation, the interaction between hydraulic fractures and pre-existing fractures, and the fluid pressure distribution in the rock matrix or fracture during the fracturing process.

Since in the fully coupled 3D model the fluid flow in the rock matrix is described by the real porous seepage, some problems that exist in the equivalent approach are avoided. Compared with the conventional method, this fully coupled 3D model also has the following advantages: it is not limited to hydraulic fracturing problems with simple geometry or only a very few natural fractures like the conventional method but can simulate the hydraulic fracturing problem with an arbitrary complex fracture network. Moreover, this fully coupled 3D model can also apply to many rock fracture and failure problems and even soil problems associated with water. It is still very time-consuming to solve large-scale problems using this approach. We will develop corresponding parallel programs using a graphics processing unit (GPU) in the future to speed up the solution of large-scale problems.

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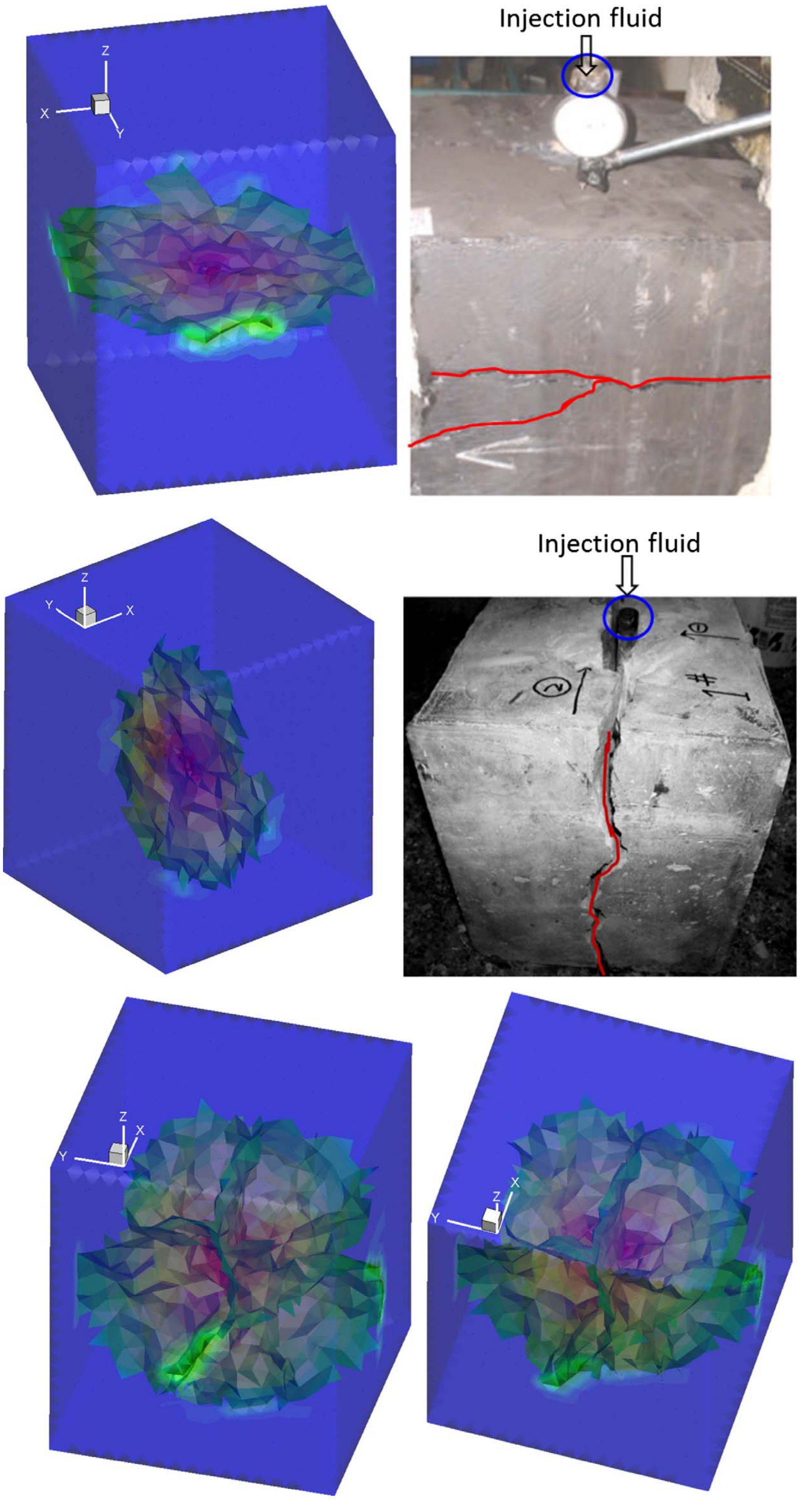

Fig. 22. Case I and II, the comparison of numerical simulation results and experimental results [78,79], and numerical crack morphology at Case III.

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